

Completing the residual-and-repair construction for benzel tilings

All deletion residues, the exceptional endpoints, and the original Split-Zero comparison

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1. Result, scope, and provenance

Constructive residue-zero theorem. For every pair of integers

$$d \geq 2, \quad 0 \leq h \leq t_d, \quad t_d = d(d-1)/2,$$

the constructor in `complete_theorem.py` produces an original type-103 tiling of

$$W_h(d) = V(d + 3h, 2d + 3h).$$

It contains exactly

$$\Delta = t_d - h \quad \text{right stones}, \quad 3h(h + d) \quad \text{bones}.$$

Reflection produces the reversed parameter pair. Every coefficient in the output is a nonnegative integer; the original incidence vector is the all-ones vector of the original region. No search is called by the constructor. All canonical deletion counts are covered, including 7, 8, 10, and 11.

This is a complete, certificate-assisted research proof of this construction, not a declaration of independent specialist acceptance. Its prerequisites are the retained full stable-packing argument, the retained threefold construction, and the retained first nonzero residue construction. Their original source files are included unchanged and their exact verifiers are replayed. The present note supplies the previously missing second residue construction and the four exceptional endpoints, and proves that the parameter coverage is exhaustive.

The original Problem 7 has two other residue families. Section 11 gives a direct construction for residue one and records the exact previously established compression-theorem input for residue two. Thus the hard residue-zero gap in the previously recorded route is closed. This package does not pretend to supply a new self-contained proof of that published compression theorem.

Provenance correction. The owner supplied a report of Akshat Sharma's earlier full-solution claim, *A Constructive Proof of Propp's Problem 7*, dated 8 September 2026, and the earlier published bone-homology calculation of Muzika Dizdarević, Timotijević, and Živaljević. Neither bare tiling existence nor the bone quotient is advertised here as new. This continuation uses the existing residual packing and original local repairs; it did not import Sharma's construction. It establishes our displayed argument, not a line-by-line audit of Sharma's manuscript. Independent review and a new Lean build have not been performed.

2. Original cells and inverse parameter maps

The axial cell is $(x, y) \in \mathbb{Z}^2$, corresponding to the barycentric cell $(x, y, 1 - x - y)$; projection onto the first two coordinates is the inverse. Its differences are

$$u = y - x, \quad v = 1 - x - 2y, \quad w = 2x + y - 1.$$

The original benzel consists of cells with all three differences in $[1 - a, b - 1]$. The four original permitted tiles are

Kind	Cell offsets at its original anchor
Right stone R	$(0, 0), (1, 0), (0, 1)$
Horizontal bone H	$(0, 0), (1, 0), (2, 0)$
Vertical bone V	$(0, 0), (0, 1), (0, 2)$
Diagonal bone D	$(0, 0), (1, -1), (2, -2)$

The differential ∂ sends each placed tile generator to the sum of its three original cell generators. The same map on free nonnegative monoids sends actual tilings to their original coverage vectors. The inclusion of these monoids in their free integer group completions commutes with ∂ ; no integer chain is called a tiling merely because it has the same image.

Use the original affine rotation and reflection

$$\rho(x, y) = (y, 1 - x - y), \quad F(x, y) = (x, 1 - x - y).$$

Their inverses are ρ^2 and F . On the original differences,

$$(u, v, w) \circ \rho = (v, w, u), \quad (u, v, w) \circ F = (-w, -v, -u).$$

Consequently ρ preserves the original benzel, and F maps $V(a, b)$ bijectively to $V(b, a)$. It preserves the right stone and permutes the three bones. Each induced tile map and cell map commutes with incidence by summing the same three cell images.

For the residue-zero family the original parameter maps are

$$(d, h) \mapsto (a, b) = (d + 3h, 2d + 3h), \quad d = b - a, \quad h = (2a - b)/3.$$

These are inverse on the stated integer lane. The right-minus-left stone invariant becomes $\Delta = t_d - h$. The region contains

$$|W_h(d)| = 3t_d + (9d - 3)h + 9h^2 = 3\Delta + 9h(h + d)$$

original cells. The retained packing proof derives this count on the original carrier; the finite verifier also reconstructs regions through the integer inverse

$$x = (1 - v - 2u)/3, \quad y = x + u,$$

retaining its divisibility condition.

3. The canonical finite core and its receiving maps

For $\Delta > 0$, let m be the unique integer with

$$t_{m-1} < \Delta \leq t_m,$$

and set $q = t_m - \Delta$. Then

$$0 \leq q \leq m - 2, \quad m \leq d, \quad h = t_d - t_m + q.$$

Conversely, these equations recover the original Δ and h . Uniqueness follows from strict increase of the triangular numbers from $t_1 = 0$. For $\Delta = 0$, use $m = 1, q = 0$. These choices are implemented by an integer square-root inverse, with its defining inequalities checked.

The full stable-packing proof is retained at `../turn8/sources/turn6-PROOF.md`. Its original rank representatives are

$$p_{r,s} = (r - s, -s), \quad 0 \leq s < r, \quad \text{rank}(p_{r,s}) = t_r + s + 1,$$

with their three rotated copies. Locating the unique minimum among the three differences and then its negative value r and coordinate s gives the inverse cell map. The first Δ rank orbits form the actual residual Ω_Δ .

The retained positive band family $\Gamma_\Delta(d, h)$ obeys

$$\partial\Gamma_\Delta(d, h) + \mathbf{1}_{\Omega_\Delta} = \mathbf{1}_{W_h(d)}. \quad (1)$$

Every band is made of original bones, and distinct bands and sectors are disjoint. Its exact coordinate proof is included, rather than only a reference to an unavailable construction. For triangular $\Delta = t_m$,

$$\Omega_{t_m} = W_0(m),$$

and the explicit all-stone base is

$$T_0(m) = \{R(m - 2 - 2r - s, r - s) : r, s \geq 0, r + s \leq m - 2\}. \quad (2)$$

For $m = 1$, this is empty. Equations (1) and (2) prove the entire $q = 0$ family.

For $q \geq 1$, retain the common original source bones

$$a_{p,y,\ell} = F\rho^p H(y - r_y - 2 - 3\ell, y), \quad r_y = \begin{cases} m - 1 & y \leq q, \\ m & y > q, \end{cases} \quad (3)$$

where

$$p \in \{0, 1, 2\}, \quad 1 \leq y \leq m + q, \quad 0 \leq \ell < \min(y, q).$$

Call this complete matching A . It is disjoint from Ω_Δ .

At $d = m, h = q$. A prefix bone has index $j = y - 1 - \ell$. In a tail row, the original first anchor is $y - 3q - m + 1$, and index $j = q - 1 - \ell$ gives exactly $y - m - 2 - 3\ell$. The index maps are

$$j_{\text{tail}} = j_{\text{stable}} + q - y, \quad j_{\text{stable}} = j_{\text{tail}} + y - q. \quad (4)$$

All original index bounds are retained. For $d \geq m + 1, h \geq m + q$, so the same source bones all belong to the finite prefix. Equations (3) and (4) preserve the sector, anchor, orientation, and three cells; hence they commute with the original differential.

A positive core tiling of $\Omega_\Delta \sqcup \text{cells}(A)$ extends to the original benzel by adjoining the actual frozen bones $\Gamma_\Delta \setminus A$. Deleting those frozen bones is the inverse on tilings containing them. This is the positive-fibre map used below; the smallest exterior parameter has not been omitted.

4. The uniform second-residue scaffold

In this section

$$k \geq 4, \quad m \geq 3k + 4, \quad q = 3k + 2.$$

The same scaffold also admits the fixed specializations $k = 2, 3$ in Section 7.

Let $B_8(m, k)$ and $S_8(m, k)$ be the exact replacement bones and retained base stones of the completed $q = 3k$ construction, whose full sources and symbolic verifier are preserved in `../turn8`. Remove from S_8 the two original stones

$$R(m - 2 - k, -k), \quad \rho^2 R(m - 2 - k, -k), \quad (5)$$

and translate the surviving stones and B_8 by $(-1, -1)$. Denote the resulting families by S and B_0 .

The two stones in (5) have base indices $(r, s) = (0, k)$ and $(m - k - 2, 0)$. The domain gives $m - k - 2 \geq 2k + 2$; they are distinct and survive the earlier corner deletions. Their removal leaves exactly $t_m - 3k - 2$ stones. This choice is essential: deleting the first and the once-rotated stone instead leaves the vacancy in the wrong place for the second channel.

Define $A_0 \subset A$ by the following original band-label table. All endpoints are included. An empty depth interval contributes no generators.

Sector	Inclusive depth interval	Inclusive row interval
0	0	$1 \leq y \leq m + 2k$
0	$1 \leq \ell \leq k - 1$	$\ell + 1 \leq y \leq m + 2k - 2$
0	$1 \leq \ell \leq k - 1$	$m + 2k + \ell - 1 \leq y \leq m + 2k + \ell$
0	k	$k + 1 \leq y \leq m + 3k$
0	$k + 1 \leq \ell \leq 2k - 1$	$\ell + 1 \leq y \leq 2k + 2$
0	$2k$	$y = 2k + 1$
1	$0 \leq \ell \leq k - 2$	$\ell + 1 \leq y \leq m + 2k - 5$
1	$0 \leq \ell \leq k - 2$	$m + 2k + \ell - 3 \leq y \leq m + 2k + \ell - 2$
1	$k - 1$	$k \leq y \leq m + 3k - 3$
1	k	$k + 1 \leq y \leq 2k - 1$
1	k	$3k \leq y \leq 3k + 2$
1	$k + 1 \leq \ell \leq 2k - 2$	$\ell + 1 \leq y \leq 2k - 1$
2	$0 \leq \ell \leq k - 3$	$\ell + 1 \leq y \leq m + 2k - 5$
2	$0 \leq \ell \leq k - 3$	$m + 2k + \ell - 2 \leq y \leq m + 2k + \ell - 1$
2	$k - 2$	$k - 1 \leq y \leq m + 3k - 3$
2	$k - 1$	$k \leq y \leq 2k - 1$
2	$k - 1$	$3k \leq y \leq 3k + 2$
2	$k \leq \ell \leq 2k - 3$	$\ell + 1 \leq y \leq 2k - 1$

The table is literal, with no chosen quotient coordinates replacing (p, y, ℓ) . Within each sector its overlapping depth ranges have disjoint row intervals. All labels obey (3). `source_domains.py` reconstructs and exactly verifies the nonnegative lengths, membership bounds, branch bounds, and pairwise distinctness over the complete parameter domains.

Write the eighteen cells of the preceding first-residue scaffold as the disjoint union

$$Z_9 = A_9 \sqcup F_9 \sqcup R_9 \sqcup U_9,$$

where

$$\begin{aligned} A_9 &= \alpha(2k-2, 4-4k-m), & \alpha(x, y) &= \{(x, y), (x, y+1), (x+1, y-1)\}, \\ F_9 &= \{(3k-1, 1-m), (3k-1, 2-m), (3k, 2-m), \\ &\quad (3k, 3-m), (3k+1, 3-m), (3k+1, 4-m)\}, \\ R_9 &= R(m-k-4, 1-k) \end{aligned}$$

as an original three-cell set, and

$$U_9 = V(m+2k-5, 2k-4) \sqcup \{(m+2k-2, 2k-1), (m+2k-2, 2k), (m+2k-1, 2k)\}.$$

Here U_9 has six cells. Define the actual affine maps

$$T_1(x, y) = (x+1, y-2), \quad T_2(x, y) = \rho^2(x, y) + (-2, 1).$$

Their inverses are translation by $(-1, 2)$ and $z \mapsto \rho(z - (-2, 1))$. Their action on placed tiles is obtained by applying the map to the three original cells.

The new 27-cell residual is

$$Z_2 = T_1(A_9 \sqcup F_9 \sqcup R_9) \sqcup T_2(F_9 \sqcup R_9 \sqcup U_9). \quad (6)$$

The exact scaffold identity is

$$\partial(B_0 + S) + \mathbf{1}_{Z_2} = \mathbf{1}_{\Omega_\Delta} + \partial A_0. \quad (7)$$

`source10.py` expresses (7) in six independent Laurent variables. Its original integer numerator vanishes identically, as detailed in Section 8. The right side is the indicator of a disjoint union by (3); every term on the left is nonnegative. Thus the identity also proves that (6) is a disjoint 27-cell set and that $B_0 + S$ is a positive packing of exactly its stated complement. Adding $A \setminus A_0$ gives the actual initial scaffold on the full common support.

5. The eight-bone bridge and the completed endpoint identity

The parent first-residue edit sequence is an explicit signed bone vector E_9 , with

$$\partial E_9 = \mathbf{1}_{Z_9}.$$

Its last upper patch, call it U , has boundary

$$\partial U = \mathbf{1}_{U_9} + \mathbf{1}_{\beta(c-1, b')}, \quad c = m+2k-4, \quad b' = 2k-5,$$

where

$$\beta(x, y) = \{(x, y), (x+1, y), (x+1, y+1)\}.$$

This equation is the original eighteen-cell terminal partition in `../turn9/PROOF.md`, not an abstract endpoint assertion.

The new bridge has anchor

$$x = m+2k-4, \quad y = 2k-7.$$

Its old and new bones, in coordinates relative to (x, y) , are:

Old bones	New bones
$H(0, 4), H(0, 5), H(0, 6)$	$V(0, 0), V(0, 3), V(1, 0)$
$H(1, 2), H(1, 3), H(1, 7)$	$V(1, 3), V(2, 2), V(2, 5)$
$V(0, 1), V(3, 4)$	$V(3, 2), V(3, 5)$

Let B_6 denote new minus old. Expanding the original offsets shows that both the old bones plus $\beta(x, y)$, and the new bones plus $\beta(x, y + 6)$, partition the same 27 cells. Consequently

$$\partial B_6 = \mathbf{1}_{\beta(x, y)} - \mathbf{1}_{\beta(x, y+6)}. \quad (8)$$

The relative coordinate map is translation by (x, y) , with translation by $(-x, -y)$ as inverse. The two positive partitions give inverse local replacement maps.

The original endpoint coordinates satisfy

$$T_1\beta(c - 1, b') = \beta(x, y), \quad T_2A_9 = \beta(x, y + 6). \quad (9)$$

Define the complete second-residue edit chain

$$E_2 = T_1(E_9 - U) + B_6 + T_2E_9. \quad (10)$$

Substitute the three original boundaries and use (9):

$$\begin{aligned} \partial E_2 &= \mathbf{1}_{T_1(A_9 \sqcup F_9 \sqcup R_9)} - \mathbf{1}_{T_1\beta(c-1, b')} \\ &\quad + \mathbf{1}_{\beta(x, y)} - \mathbf{1}_{T_2A_9} + \mathbf{1}_{T_2Z_9} \\ &= \mathbf{1}_{Z_2}. \end{aligned} \quad (11)$$

This is the endpoint cancellation that the interrupted attempt lacked. Its two geometric changes are the eight-bone bridge and the second deleted stone in (5).

The formula is also expanded independently by `symbolic_certificate.py`. It releases $4m + 12k + 1$ old scaffold bones and inserts $4m + 12k + 10$ new bones, a gain of nine bones, exactly the 27 filled cells. These are counts of the displayed repair, not an optimality assertion.

6. Original source ownership and positivity

A signed identity does not supply the old generators needed to execute a positive edit. For every negative generator in (10), `edits.json` records an original owner of one of two types:

1. a common source label (p, y, ℓ) from (3), proved to lie outside A_0 ;
2. an exact family and index in the original parent $B_8(m, k)$, followed by its actual rotation and the initial translation $(-1, -1)$.

These are literal inverse index formulas. `ownership.py` substitutes them into all three original cell coordinates and verifies equality as an identity in (m, k, i) . Bounds are proved on the full parameter domain. Every distinct pair of atoms that could share an owner is checked against the two original label coordinates; their equality produces a rational contradiction. Within a family, a nonzero index coefficient proves injectivity.

The verifier reconstructs 1,479 exact affine certificates for the uniform second residue, including 71 original generator identities and 733 potential duplicate-source comparisons. It uses only Python

integer and rational arithmetic. A certificate for a bound is a nonnegative rational combination of the domain inequalities equal to that bound. A contradiction certificate combines the domain and equality constraints into $-1 \geq 0$. No floating-point feasibility status is accepted as a certificate.

The initial scaffold is a matching by (7), and the source certificate proves that all negative edit generators are distinct members of that initial scaffold. Subtracting them therefore leaves nonnegative coefficients. Adding the positive edit generators leaves a nonnegative tile vector. Equations (7) and (11) give its original boundary:

$$\partial(B' + S) = \mathbf{1}_{\Omega_\Delta} + \partial A. \quad (12)$$

The target coefficients are all one, so every tile is contained, all tiles are disjoint, and every target cell is covered. This proves the positive core tiling throughout the unbounded domain. No step assumes a positive lifting theorem.

The source-to-tail map (4) now returns the core to every original exterior benzel. Adjoining the frozen original bones gives

$$(\Gamma_\Delta(d, h) \setminus A) \cup B' \cup S, \quad (13)$$

and (1), (12) prove its positive partition. The counts are $t_m - 3k - 2$ stones and $3h(h + d)$ bones.

7. The four exceptional deletion counts

The parent terminal prescription uses $k - 4$ vertical vacancy moves and must not be used as a negative-length formula at $k = 2, 3$. The following exact constant patches replace it.

Let P be the parent edit sequence stopped immediately before its **lower** group. Set

$$c = m + 2k - 4, \quad b = 1 - k.$$

For $k = 2, 3$, the remaining twelve cells, relative to (c, b) , are

$$\begin{aligned} H_k &= \alpha(0, -3) \sqcup R(-3, 0) \sqcup V(-1, 3k - 5) \\ &\sqcup \{(2, 3k - 2), (2, 3k - 1), (3, 3k - 1)\}. \end{aligned} \quad (14)$$

The fixed terminal vector Q_k , printed in Appendix A, satisfies

$$\partial Q_k = \mathbf{1}_{H_k}.$$

For $k = 2$, it releases 11 bones and inserts 15. For $k = 3$, it releases 14 and inserts 18. Every equality is proved by expanding the literal finite offsets. Thus $P + Q_k$ completes the first residue for

$$(q, m) = (7, m \geq 9), \quad (10, m \geq 12).$$

For the second residue, apply $T_1 P$, then a fixed direct bridge D_k , then $T_2(P + Q_k)$. The direct bridge is anchored at

$$(c', b') = (m + 2k - 3, -k - 1).$$

In its relative coordinates it replaces

$$\alpha(0, -3) \sqcup R(-3, 0) \quad \text{by} \quad \beta(-1, 3k).$$

For $k = 2$, it releases 16 bones and inserts 17; for $k = 3$, it releases 20 and inserts 21. The complete tables are in Appendix A. The resulting vacancy is exactly $T_2 A_9$, because its anchor is

$$(m + 2k - 4, 2k - 1).$$

The endpoint cancellation is therefore the same original cell cancellation as (11), with the direct bridge taking the place of the invalid short lower/beta prescription.

This proves the second residue for

$$(q, m) = (8, m \geq 10), \quad (11, m \geq 13).$$

For these four branches, k is fixed but m is not bounded above. The exact symbolic verifier specializes only k , leaving the original exterior parameter formal. Source ownership is proved by the same affine mechanism on the domains $k = k_0$, $m \geq 3k_0 + r + 2$. It does not claim an identity for other k -values.

The four source verifications have, respectively, 407, 968, 559, and 1,178 exact certificates. The initial sources are retained and every old edit generator remains distinct. The positivity proof of Section 6 applies to these literal tables. The smallest finite tail is included by (4).

8. Exact algebraic certificates and index domains

The cell-module map

$$\mathbb{Z}[\mathbb{Z}^2] \longrightarrow \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}], \quad [(x, y)] \mapsto X^x Y^y$$

has inverse coefficient extraction. The four tile boundaries are exactly

$$1 + X + Y, \quad 1 + X + X^2, \quad 1 + Y + Y^2, \quad 1 + XY^{-1} + X^2 Y^{-2}.$$

Retain both independent parameters in

$$\mathcal{L} = \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}, M_x^{\pm 1}, M_y^{\pm 1}, K_x^{\pm 1}, K_y^{\pm 1}].$$

The original specialization is

$$M_x \mapsto X^m, \quad M_y \mapsto Y^m, \quad K_x \mapsto X^k, \quad K_y \mapsto Y^k.$$

Its kernel is generated by the four displayed parameter relations, and the quotient inverse is the inclusion of X, Y . This identifies the exact relationship between formal parameter calculations and original cells.

For an indexed anchor whose coordinates are affine in m, k, i , the finite sum is represented by its actual geometric progression. The checker expands constant families directly. All denominators are nonzero Laurent polynomials in X, Y alone; specialization never changes them. The original Laurent rings are integral domains, so localization at their product is injective.

For the new uniform branch, the scaffold numerator has 273 rational input terms, 32,274 expanded numerator terms, and no surviving coefficient. The endpoint identity has 228 rational input terms, 10,512 expanded terms, and no surviving coefficient. The separate fixed- k expansions for the four exceptions also have zero numerator. These are exact integer polynomial identities before choosing m ; the uniform branch also retains k .

Positive source domains are checked separately. `source_domains.py` covers both uniform scaffold tables, the four fixed specializations, and the four retained fixed-deletion tables $q = 1, 2, 4, 5$. It verifies 1,291 exact affine bounds/contradictions: nonnegative family lengths, original row/depth bounds, each prefix/tail branch, and source distinctness. The four fixed-deletion polynomial identities are replayed as well. Their construction data are preserved in `../turn7`.

This makes the proof's logical order explicit: the polynomial identity establishes original cell coefficients; the index certificates establish that the expressions enumerate actual nonnegative generators; the source owners establish that subtraction removes actual present tiles. Neither a rational-function identity alone nor a finite tiling sample is substituted for those three calculations.

9. Exhaustion of all canonical parameters

Every $q \in [0, m - 2]$ has exactly one route in the following table.

Canonical deletion count	Complete construction	Original domain
$q = 0$	Stable bands plus the all-stone core (2)	$m \geq 1$
$q = 1, 2, 4, 5$	Retained literal fixed-deletion tables	$m \geq q + 2$
$q = 3k, k \geq 1$	Retained threefold construction	$m \geq 3k + 2$
$q = 3k + 1, k \geq 4$	Retained asymmetric construction	$m \geq 3k + 3$
$q = 3k + 2, k \geq 4$	Sections 4–6	$m \geq 3k + 4$
$q = 7, 10$	Fixed $k = 2, 3$, first-residue terminal	$m \geq q + 2$
$q = 8, 11$	Fixed $k = 2, 3$, direct second-residue bridge	$m \geq q + 2$

To verify exhaustiveness, divide q by three. For remainder zero, $q = 0$ or the threefold row applies. For remainder one, quotient $k = 0, 1, 2, 3$ gives $q = 1, 4, 7, 10$, and all larger quotients use the uniform first residue. For remainder two, quotients $0, 1, 2, 3$ give $2, 5, 8, 11$, and all larger quotients use the new uniform second residue. Every required lower bound on m is exactly the already proved canonical inequality $q \leq m - 2$.

Thus no canonical value is left to a search, unproved lifting statement, or future endpoint. The original receiving parameter is $d \geq m$, so (13), or the corresponding retained literal construction, applies.

The two small $d = 2$ cases are also explicit. For $h = 0$, $V(2, 4)$ is the central right stone. For $h = 1$, the same original three-sector band formula gives the following nine bones:

$$\begin{aligned}
 &D(-2, 1), \quad D(-2, 2), \quad D(-1, -1) \\
 &H(-3, 3), \quad H(-1, 2), \quad H(0, 1) \\
 &V(1, -2), \quad V(2, -2), \quad V(3, -1)
 \end{aligned}$$

Their cell sets partition the original $V(5, 7)$, as direct offset expansion shows. This completes the theorem in Section 1 for every $d \geq 2$.

10. Original Split-Zero complexes and their comparison kernels

This section records the concepts used in the construction, not a claim that adjoining a zero alone supplies positive tilings. The source is the retained Split-Zero reconstruction and internal-homology development at the exact mathematical pin listed in `PROVENANCE.md`.

Fix one of the newly completed repair configurations. Let A^- be its matching of old edit bones and A^+ its matching of new edit bones, and let Z be its actual initial vacancy. The proved endpoint identity is

$$\partial(A^+ - A^-) = \mathbf{1}_Z. \tag{15}$$

The supports Z and the old cells are disjoint. Take

$$L = \{\perp\} \sqcup \mathcal{P}(A^-), \quad \Lambda_E = Z \sqcup \text{cells}(E).$$

Bottom has empty physical support, whereas the nonbottom label $E = \emptyset$ has physical support Z . The join map to cell supports sends unions to unions and bottom to the empty set. Its inverse on the image identifies which disjoint old bone cells are included.

At each supported label use the actual incidence complex of contained bone generators

$$C_E^0 \xrightarrow{\partial_E} C_E^1 \longrightarrow 0, \quad C_E^1 = \mathbb{Z}[\Lambda_E].$$

The original inclusions intertwine differentials. Reconstruct the totals $\mathcal{T}(C^i) = \coprod_E C_E^i$ over $G(\mathbb{Z}) = \mathbb{Z} \sqcup \{\tau\}$. Their addition transports to the union support. The supported scalar zero $e = 0_{\mathbb{Z}}$ and absent zero act by

$$e(E, x) = (E, 0), \quad \tau(E, x) = (\perp, 0).$$

The differential square is the correctly typed map $(E, n) \mapsto (E, 0)$. Projection to the support proves its relationship to the constant absent map; no label disappears when a boundary becomes zero.

The internal homology projection

$$(E, p) \longmapsto (E, [p])$$

coequalizes the actual incoming boundary and its supported-zero companion. Its universal property follows directly: a split-linear map equalizing those arrows takes two representatives differing by an original boundary to the same value, using fibre-zero absorption. It therefore descends uniquely; lifting representatives verifies addition, scalar action, and naturality. No cancellation in an arbitrary target semimodule is used.

10.1 The augmented positive fibre

Let $v_E = \mathbf{1}_{\Lambda_E}$ and

$$\widehat{C}_E^0 = \mathbb{Z} \oplus C_E^0, \quad \widehat{d}_E(r, n) = \partial n - r v_E.$$

For $E \subseteq E'$, let $\eta_{EE'}$ be the sum of newly released original bones. The original partition gives

$$v_{E'} = j v_E + \partial \eta_{EE'}.$$

The genuine linear transition

$$\widehat{j}_{EE'}(r, n) = (r, j n + r \eta_{EE'})$$

therefore satisfies $\widehat{d}_{E'} \widehat{j}_{EE'} = j \widehat{d}_E$. The identity $\eta_{EE''} = j \eta_{EE'} + \eta_{E'E''}$ proves composition.

Charge-one nonnegative integral cycles are exactly the original positive patch tilings: $\partial n = v_E$ forces coefficient one at each cell. The inclusion into the full integral cycle module retains the same charge and every original tile coefficient. Equation (15) supplies the actual positive cycle $(1, A^+)$ at full support. The elementary positive partitions and their inverses supply the concrete local maps realizing that cycle.

10.2 The full integral matching-comparison kernel

At the fixed full support, write $C = \mathbb{Z}[\Lambda]$, $B_- = \text{im } \partial_{A-}$, and $B_+ = \text{im } \partial_{A+}$. The comparison is the original representative map

$$C/B_- \longrightarrow C/(B_- + B_+),$$

with kernel

$$K \cong B_+/(B_- \cap B_+). \quad (16)$$

The map sends a new-boundary representative to its original class. Its inverse writes a killed representative as $a + b$, with $a \in B_-$, $b \in B_+$, and sends it to $[b]$. Two decompositions differ precisely by $B_- \cap B_+$, proving well-definedness and both inverse identities.

For completeness, the entire kernel is computed integrally. Build the bipartite graph of actual old and new bones, with each shared original cell as an edge. A singly owned cell marks its owner. Equality of old and new boundary combinations forces coefficients equal along edges and zero at marked vertices. They vanish on open components and have one common integer value on each closed component. Conversely, a closed component gives the equality of the sums of its original old and new boundaries.

Those intersection generators are primitive and have disjoint supports. Pick one new-bone pivot in each closed component. The remaining new-boundary classes form an integral basis of (16). The coordinate map subtracts the pivot coefficient on each closed component; the inverse inserts zero at pivots. Original-cell forest paths give integer duals: root each open component at ground and each closed component at its pivot, and assign oriented path coefficients 0, 1, -1 to its cell edges. Summing at old vertices gives zero, and summing at new vertices gives the Kronecker evaluations of the displayed basis.

Any chosen vacancy cell evaluates $[1_Z]$ to one and annihilates every old boundary. It supplies a split copy of $\mathbb{Z}$ before the comparison. Equation (15) kills that distinguished class at its own support. This one summand is retained alongside the full kernel, not substituted for it.

The selected matching complexes also map into the complex of all contained bones by literal inclusion on tile generators and identity on cells. The further quotient kernel is

$$\text{im } \partial_{\text{all}}/(B_- + B_+).$$

This records precisely the relationship with the unrestricted incidence complex; no vanishing of that additional kernel is assumed. None of these elementary matching calculations is claimed to be a new computation of the known global bone homology group.

11. Returning to the original Problem 7

The retained source-matching note `reference-earlier-proofs.md`, Section 3, specifies the original admissible domain $a, b \geq 2$, $a \leq 2b$, $b \leq 2a$, and the three residue cases of the Conway–Lagarias invariant. That note is included unchanged and its historical status statements remain historical. Reflection permits $a \leq b$ through the exact involution in Section 2.

For $a + b \equiv 0 \pmod{3}$, the inverse parameters are $d = b - a$, $h = (2a - b)/3$, and $\Delta = t_d - h$. Admissibility gives $h \geq 0$, and nonnegative invariant gives $h \leq t_d$. The possibilities $d = 0, 1$ would force $h = 0$ and violate $a \geq 2$. Thus $d \geq 2$, and the completed constructor applies.

For $a + b \equiv 1 \pmod{3}$, there is a direct all-right-stone construction. Put $A = 1 - a$, $B = b - 1$, $r = 2 - a \pmod{3}$. For an original barycentric cell v , consider the three sum-zero anchors $u = v - e_j$ of containing

right stones. Their first differences occupy all three residues; choose the unique anchor in residue r . All differences of a sum-zero anchor are congruent modulo three. They lie initially in $[A - 1, B + 1]$, but the endpoints $A + 1, B - 1$ both have residue r , and none of $A - 1, A, B, B + 1$ does. Hence the chosen anchor differences lie in $[A + 1, B - 1]$, and its three cells lie in the original benzel. Every cell of that stone selects the same anchor. The map from selected anchors and offset indices to original cells is therefore bijective, with the displayed selection as inverse. `residue_one` implements this construction directly.

For $a + b \equiv 2 \pmod{3}$, the retained source-matching note invokes the already published generalized-compression Theorem 1.1 at arXiv:2403.07663v1, in the exact parameters

$$n = b + 1 - a, \quad j = (2a - b - 1)/3, \quad (a, b) = (n + 3j, 2n + 3j - 1).$$

The inverse formulas are displayed and $n \geq 1, j \geq 0$ follow from admissibility and the residue. That established result gives positive tiling counts using the original right stone and two original bone orientations. The literal inclusion of its tile set into the type-103 set preserves each placement and its three cells; its induced incidence square commutes and its map on tilings is injective. This is the exact published input of the full-P7 assembly. The present package does not assert a self-contained new construction or independent proof audit of that compression result.

Accordingly, the previously recorded route to the full existence statement now has its remaining residue-zero case supplied by a complete original-coordinate construction. This is independent of Sharma’s reported manuscript in method and proof dependencies, while making no claim of first proof or novel bare existence.

12. Verification, limitations, and reproduction

The mathematical proof uses parameter-preserving polynomial identities, exact affine certificates, the retained full packing proofs, and the positive incidence arguments above. Finite tests are additional checks of the implementation, not the source of an unbounded quantifier.

Run from this directory:

```
python verify.py --max-d 12 --max-k 9 --output evidence/normal.json
python -O verify.py --max-d 12 --max-k 9 --output evidence/optimized.json
```

The verifier uses only the Python standard library. Optional scripts that discovered rational multipliers or fixed finite patches are not invoked. The source map, all small constant tables, old-generator inverse maps, exact certificates, and retained mathematical dependencies are included.

A complete run checks every allowed h at $d = 2, \dots, 12$, its reflection, larger uniform second-residue cores at their smallest and larger m , all four exceptional families at three m -values and two receiving values, all duals in eight original matching comparisons, actual augmented transitions, and deliberately corrupted inputs. Exact counts and file identities belong to the emitted receipts.

There is no first-discovery claim, no new Lean build certificate, and no independent specialist acceptance. This manuscript closes the displayed construction and its canonical parameter coverage. The separate prior literature claims and the published residue-two input keep their own provenance and verification scopes.

Appendix A. The four constant terminal tables

All coordinates in these tables are relative to the anchors in Section 7. The table entries are original placed bones, not quotient representatives. Old/new equalities follow by literal offset expansion; `small_symbolic.py` verifies the aggregate unbounded- m identities and the ownership certificates supply the actual source indices.

Terminal Q_2

Original releases: 11 bones. Original insertions: 15 bones.

Old placed bone	New placed bone
$H(-1, 0)$	$D(0, -3)$
$H(0, 1)$	$D(0, -2)$
$H(0, 2)$	$D(0, 0)$
$H(0, 3)$	$D(0, 1)$
$V(-2, 1)$	$D(1, -2)$
$V(1, -3)$	$H(-3, 0)$
$V(2, -5)$	$H(-3, 1)$
$V(2, -2)$	$H(-2, 2)$
$V(3, -4)$	$H(-2, 3)$
$V(3, -1)$	$V(1, 1)$
$V(3, 2)$	$V(2, 0)$
	$V(2, 3)$
	$V(3, -3)$
	$V(3, 0)$
	$V(3, 3)$

Direct bridge D_2

Original releases: 16 bones. Original insertions: 17 bones.

Old placed bone	New placed bone
$H(-1, 0)$	$D(0, -3)$
$H(-1, 6)$	$D(0, -2)$
$H(0, 1)$	$D(0, 0)$
$H(0, 2)$	$D(0, 1)$
$H(0, 3)$	$D(1, -2)$
$H(0, 7)$	$H(-3, 0)$
$V(-2, 1)$	$H(-3, 1)$
$V(-1, 1)$	$H(-2, 2)$
$V(1, -3)$	$H(-2, 3)$
$V(2, -5)$	$H(1, 6)$
$V(2, -2)$	$H(1, 7)$
$V(2, 4)$	$V(1, 1)$
$V(3, -4)$	$V(2, 0)$
$V(3, -1)$	$V(2, 3)$
$V(3, 2)$	$V(3, -3)$
$V(3, 5)$	$V(3, 0)$
	$V(3, 3)$

Terminal Q_3

Original releases: 14 bones. Original insertions: 18 bones.

Old placed bone	New placed bone
$H(-1, 0)$	$D(0, -3)$
$H(1, 4)$	$D(0, -2)$
$V(-2, 1)$	$D(1, -2)$
$V(-1, 1)$	$D(1, -1)$
$V(0, 2)$	$D(2, -1)$
$V(1, -3)$	$H(-3, 0)$
$V(2, -5)$	$H(-3, 1)$
$V(2, -2)$	$H(-2, 2)$
$V(3, -4)$	$H(-2, 3)$
$V(3, 5)$	$H(0, 0)$
$V(4, -3)$	$H(0, 4)$
$V(4, 0)$	$H(2, 7)$
$V(4, 3)$	$H(2, 8)$
$V(4, 6)$	$V(-1, 4)$
	$V(3, 4)$
	$V(4, -2)$
	$V(4, 1)$
	$V(4, 4)$

Direct bridge D_3

Original releases: 20 bones. Original insertions: 21 bones.

Old placed bone	New placed bone
$H(-1, 0)$	$D(0, -3)$
$H(-1, 9)$	$D(0, -2)$
$H(0, 1)$	$D(0, 0)$
$H(0, 5)$	$D(0, 1)$
$H(0, 6)$	$D(1, -2)$
$H(0, 10)$	$H(-3, 0)$
$H(1, 2)$	$H(-3, 1)$
$H(1, 3)$	$H(-2, 2)$
$H(1, 4)$	$H(-2, 3)$
$V(-2, 1)$	$H(1, 9)$
$V(-1, 1)$	$H(1, 10)$
$V(0, 2)$	$V(0, 4)$
$V(1, -3)$	$V(1, 1)$
$V(2, -5)$	$V(1, 4)$
$V(2, -2)$	$V(2, 0)$
$V(2, 7)$	$V(2, 3)$
$V(3, -4)$	$V(2, 6)$
$V(3, -1)$	$V(3, -3)$
$V(3, 5)$	$V(3, 0)$
$V(3, 8)$	$V(3, 3)$
	$V(3, 6)$

Appendix B. Literal indexed edit tables

`edits.json` contains the complete uniform second-residue table. The four exceptional tables have filenames `edits-kK-rR.json`, with $(K, R) = (2, 1), (2, 2), (3, 1), (3, 2)$. They retain all their original owner maps after fixing k . No unspecified geometric pattern is called by the constructor.

A row stores a kind and two vectors $[a, b, c, e]$; its anchor coordinate is exactly $am + bk + ci + e$. A count vector $[a, b, e]$ means $0 \leq i < am + bk + e$. Common owners store their original sector, row, depth, and triangular branch. Parent owners store the exact part, family, inverse indices, and rotation. Their equality is verified on all three original cell coordinates.

The retained full first-residue edit table gives the literal meaning of E_9 and U in (10). `build_edits.py` records their exact affine transport; the final constructor reads the fully expanded table and calls no discovery script.